Digital System Design and Implementation

Final Exam(111/05/31)

Spring 2022

Total points: 100 points

- Please put all your modules in **one Verilog file** named by DSDXXXXXXXXX.v or DSXDXXXXXXXX.sv, where X is your student ID, for example, DSD105501001.v.
- Ensure your module and input as well as output ports having **the same module/port names** as described in the following.
- **Open Book!!!!**

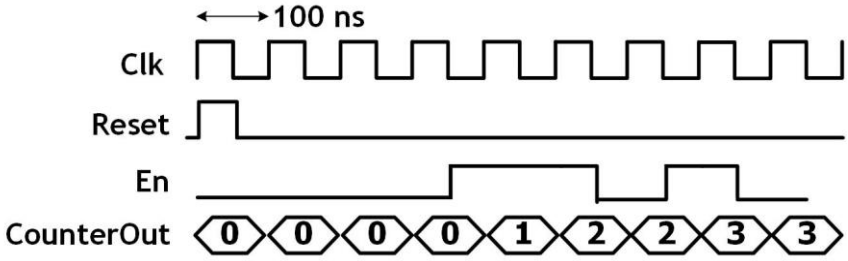
In this final exam, we want to test one sequence with specific patterns. Please write each module first.

1. Please write a counter with asynchronous reset and positive edge trigger. The counter will start to count from "1" to "1023" and back to "1" only when the "**En**" signal is high. (10%)

   module EnCounter(CounterOut, Clk, Reset, En);

   ...

   endmodule



2. Please write a module called "**Rotator**" that will rotate the 4-bit input "**RIn"** to the left by r bits, where $r = 0,1,2,3$, and is set by 2-bit input signal "**Rb"**. The output is called "**ROut"**. (10%)

   module Rotator(Rout, RIn, Rb);

   ...

   endmodule

3. Please write a synchronous read/write register file that has 8 words and each word has 10 bits wide. The write enable signal "**WE**" is active high. (10%)

   module RegisterFile(Q, D, Clk, WE, Add);

   ...

   endmodule

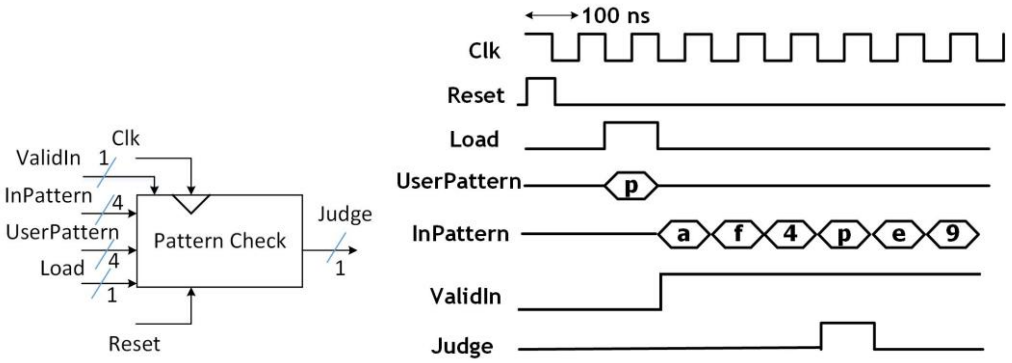
4. Please *write* a module called "**PatternCheck**" that will generate "1" at exactly the same clock time when 4-bit input pattern is matched with a pre-defined 4-bit pattern specified by the user using "**Load**" signal. The input pattern is sent by "**InPattern**" indicated by "**ValidIn**" signal that will be activated only after loading operation of the user pattern as shown in the following, where

$0 \leq p \leq 15$. (You don't need to write test pattern.) (15%)

module PatternCheck(Judge, InPattern, ValidIn, UserPattern, Load, Clk, Reset);

…

endmodule

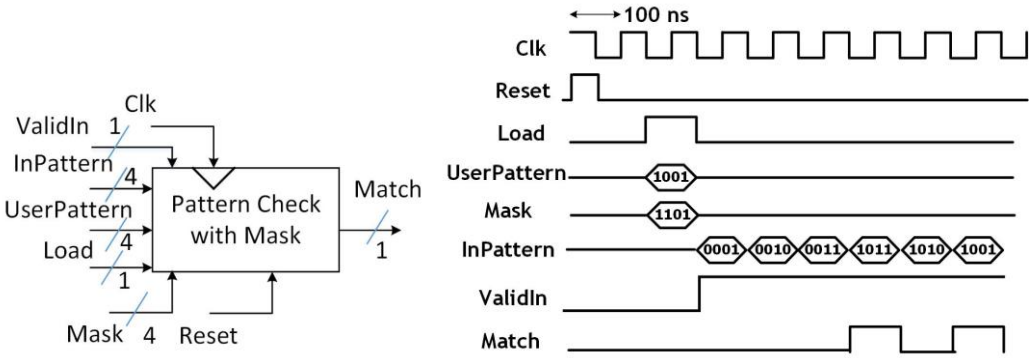


5. Please write a module called "MaskPatternCheck" that will generate "1" at exactly the same time when some bits of the 4-bit input pattern are matched with a pre-defined pattern specified by the user using "Load" signal. For example, we can specify pattern to be compared as "4'b10?1". It means that the mask is "4'b1101" and we don't care the state of the third bit. If the input pattern is "4'b1001" or "4'b1011", the output flag "Match" will become "1". (You can reuse your codes of Q4) (15%)

   module MaskPatternCheck(Match, InPattern, ValidIn, Mask, UserPattern, Load, Clk, Reset);

   …

   endmodule



6. Now, a 4-bit sequence with sequence length smaller than 1023 is going to be processed. We need to check how many times the user pre-defined pattern occurs and save the result in the address 4 of register file. If "**ValidIn**"=0, the associated "**InPattern**" is invalid and it is not counted in. Please use the template in the following and name the register file as U1 as described in the following. (In addition to Register file, you can use previous modules freely.) (20%)

   module PatternCount((InPattern, ValidIn, UserPattern, Load, Clk, Reset);

   …

   RegisterFile U1(.Q( ), .D( ), .Clk( ), .WE( ), .Add( ));

endmodule

7. Now, a 4-bit sequence with sequence length smaller than 1023 is going to be processed. We need to check the positions of the pattern "1?01" with mask "1011" occurs and save the position result of the last occurrence of that pattern in the address 7 of register file. The starting position index 1 corresponds the first data indicated by "**ValidIn"**. If "**ValidIn**"=0, the associated "**InPattern**" is invalid and its position will not be saved. Please use the template in the following and name the register file as U1 as described in the following. (In addition to Register file, you can use previous modules freely.) (20%)
   module PatternPosition((InPattern, ValidIn, Clk, Reset);
   …
   RegisterFile U1(.Q( ), .D( ), .Clk( ), .WE( ), .Add( ));
   endmodule